Machine Learning Interview Questions

What do you understand by Machine learning?

Machine learning is the form of Artificial Intelligence that deals with system programming and automates data analysis to enable computers to learn and act through experiences without being explicitly programmed.

Differentiate between inductive learning and deductive learning?

In inductive learning, the model learns by examples from a set of observed instances to draw a generalized conclusion. On the other side, in deductive learning, the model first applies the conclusion, and then the conclusion is drawn.

Inductive learning is the method of using observations to draw conclusions.

Deductive learning is the method of using conclusions to form observations.

What is the difference between Data Mining and Machine Learning?

Data mining can be described as the process in which the structured data tries to abstract knowledge or interesting unknown patterns. During this process, machine learning algorithms are used.

Machine learning represents the study, design, and development of the algorithms which provide the ability to the processors to learn without being explicitly programmed.

Uplatz

Machine Learning Interview Questions

What is the meaning of Overfitting in Machine learning?

Overfitting can be seen in machine learning when a statistical model describes random error or noise instead of the underlying relationship. Overfitting is usually observed when a model is excessively complex. It happens because of having too many parameters concerning the number of training data types.

Why overfitting occurs?

The possibility of overfitting occurs when the criteria used for training the model is not as per the criteria used to judge the efficiency of a model.

What is the method to avoid overfitting?

Overfitting occurs when we have a small dataset, and a model is trying to learn from it. By using a large amount of data, overfitting can be avoided. But if we have a small database and are forced to build a model based on that, then we can use a technique known as cross-validation. In this method, a model is usually given a dataset of a known data on which training data set is run and dataset of unknown data against which the model is tested.

Differentiate supervised and unsupervised machine learning.

In supervised machine learning, the machine is trained using labeled data. Then a new dataset is given into the learning model so that the algorithm provides a positive outcome by analyzing the labeled data. For example, we first require to label the data which is necessary to train the model while performing classification./li>

In the unsupervised machine learning, the machine is not trained using labeled data and let the algorithms make the decisions without any corresponding output variables.

Uplatz

Machine Learning Interview Questions

How does Machine Learning differ from Deep Learning?

Machine learning is all about algorithms which are used to parse data, learn from that data, and then apply whatever they have learned to make informed decisions.

Deep learning is a part of machine learning, which is inspired by the structure of the human brain and is particularly useful in feature detection.

How is KNN different from k-means?

KNN or K nearest neighbors is a supervised algorithm which is used for classification purpose. In KNN, a test sample is given as the class of the majority of its nearest neighbors. On the other side, K-means is an unsupervised algorithm which is mainly used for clustering. In k-means clustering, it needs a set of unlabeled points and a threshold only. The algorithm further takes unlabeled data and learns how to cluster it into groups by computing the mean of the distance between different unlabeled points.

What are the different types of Algorithm methods in Machine Learning?

Supervised Learning

Semi-supervised Learning

Unsupervised Learning

Transduction

Reinforcement Learning

Uplatz

Machine Learning Interview Questions

How do classification and regression differ?

Classification is the task to predict a discrete class label.

Regression is the task to predict a continuous quantity.

In a classification problem, data is labeled into one of two or more classes.

A regression problem needs the prediction of a quantity.

A classification having problem with two classes is called binary classification, and more than two classes is called multi-class classification

A regression problem containing multiple input variables is called a multivariate regression problem.

What are the five popular algorithms we use in Machine Learning?

Decision Trees

Probabilistic Networks

Neural Networks

Support Vector Machines

Nearest Neighbor

Uplatz

Machine Learning Interview Questions

What is a model selection in Machine Learning?

The process of choosing models among diverse mathematical models, which are used to define the same data is known as Model Selection. Model learning is applied to the fields of statistics, data mining, and machine learning.

What are the three stages of building the hypotheses or model in machine learning?

Model building

It chooses a suitable algorithm for the model and trains it according to the requirement of the problem.

Applying the model

It is responsible for checking the accuracy of the model through the test data.

Model testing

It performs the required changes after testing and apply the final model.

What according to you, is the standard approach to supervised learning?

In supervised learning, the standard approach is to split the set of example into the training set and the test.

Uplatz

Machine Learning Interview Questions

What are the common ways to handle missing data in a dataset?

Missing data is one of the standard factors while working with data and handling. It is considered as one of the greatest challenges faced by the data analysts. There are many ways one can impute the missing values. Some of the common methods to handle missing data in datasets can be defined as deleting the rows, replacing with mean/median/mode, predicting the missing values, assigning a unique category, using algorithms that support missing values, etc.

What do you understand by ILP?

ILP stands for Inductive Logic Programming. It is a part of machine learning which uses logic programming. It aims at searching patterns in data which can be used to build predictive models. In this process, the logic programs are assumed as a hypothesis.

What are the necessary steps involved in Machine Learning Project?

There are several essential steps we must follow to achieve a good working model while doing a Machine Learning Project. Those steps may include parameter tuning, data preparation, data collection, training the model, model evaluation, and prediction, etc.

Describe Precision and Recall?

Precision can be said as a positive predictive value. It is the fraction of relevant instances among the received instances.

On the other side, recall is the fraction of relevant instances that have been retrieved over the total amount or relevant instances. The recall is also known as sensitivity.

Uplatz

Machine Learning Interview Questions

What are the functions of Supervised Learning?

Classification

Speech Recognition

Regression

Predict Time Series

Annotate Strings

What are the functions of Unsupervised Learning?

Finding clusters of the data

Finding low-dimensional representations of the data

Finding interesting directions in data

Finding novel observations/ database cleaning

What do you understand by algorithm independent machine learning?

Algorithm independent machine learning can be defined as machine learning, where mathematical foundations are independent of any particular classifier or learning algorithm.

Uplatz

Machine Learning Interview Questions

How will you explain a linked list and an array?

ARRAY LINKED LIST

An array is a group of elements of a similar data type.

Linked List is an ordered group of elements of the same type, which are connected using pointers.

Elements are stored consecutively in the memory.

New elements can be stored anywhere in memory.

An Array supports Random Access. It means that the elements can be accessed directly using their index value, like arr[0] for 1st element, arr[5] for 6th element, etc.

As a result, accessing elements in an array is fast with constant time complexity of O(1).

Linked List supports Sequential Access. It means that we have to traverse the complete linked list, up to that element sequentially which element/node we want to access in a linked list.

Uplatz

Machine Learning Interview Questions

What do you understand by the Confusion Matrix?

TN= True Negative

TP= True Positive

FN= False Negative

FP= False Positive

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

Explain True Positive, True Negative, False Positive, and False Negative in Confusion

True Positive

When a model correctly predicts the positive class, it is said to be a true positive.

True Negative

When a model correctly predicts the negative class, it is said to be a true negative.

False Positive

When a model incorrectly predicts the positive class, it is said to be a false positive. It is also known as 'Type I' error.

False Negative

When a model incorrectly predicts the negative class, it is said to be a false negative. It is also known as 'Type II' error.

Machine Learning Interview Questions

What according to you, is more important between model accuracy and model performance?

Model accuracy is a subset of model performance. The accuracy of the model is directly proportional to the performance of the model. Thus, better the performance of the model, more accurate are the predictions.

What is Bagging and Boosting?

Bagging is a process in ensemble learning which is used for improving unstable estimation or classification schemes.

Boosting methods are used sequentially to reduce the bias of the combined model.

What are the similarities and differences between bagging and boosting in Machine Learning?

Both are the ensemble methods to get N learns from 1 learner.

Both generate several training data sets with random sampling.

Both generate the final result by taking the average of N learners.

Both reduce variance and provide higher scalability.

Differences between Bagging and Boosting

Uplatz

Machine Learning Interview Questions

What do you understand by Cluster Sampling?

Cluster Sampling is a process of randomly selecting intact groups within a defined population, sharing similar characteristics. Cluster sample is a probability where each sampling unit is a collection or cluster of elements.

What do you know about Bayesian Networks?

Bayesian Networks also referred to as 'belief networks' or 'casual networks', are used to represent the graphical model for probability relationship among a set of variables.

Which are the two components of Bayesian logic program?

Logical

It contains a set of Bayesian Clauses, which capture the qualitative structure of the domain.

Quantitative

It is used to encode quantitative information about the domain.

Uplatz

Machine Learning Interview Questions

Why instance-based learning algorithm sometimes referred to as Lazy learning algorithm?

In machine learning, lazy learning can be described as a method where induction and generalization processes are delayed until classification is performed. Because of the same property, an instance-based learning algorithm is sometimes called lazy learning algorithm.

What do you understand by the F1 score?

The F1 score represents the measurement of a model's performance. It is referred to as a weighted average of the precision and recall of a model. The results tending to 1 are considered as the best, and those tending to 0 are the worst.

How is a decision tree pruned?

Pruning is said to occur in decision trees when the branches which may consist of weak predictive power are removed to reduce the complexity of the model and increase the predictive accuracy of a decision tree model. Pruning can occur bottom-up and top-down, with approaches such as reduced error pruning and cost complexity pruning.

Uplatz

Machine Learning Interview Questions

What do you understand by Underfitting?

Underfitting is an issue when we have a low error in both the training set and the testing set. Few algorithms work better for interpretations but fail for better predictions.

When does regularization become necessary in Machine Learning?

Regularization is necessary whenever the model begins to overfit/ underfit. It is a cost term for bringing in more features with the objective function. Hence, it tries to push the coefficients for many variables to zero and reduce cost term. It helps to reduce model complexity so that the model can become better at predicting (generalizing).

What is Regularization? What kind of problems does regularization solve?

A regularization is a form of regression, which constrains/ regularizes or shrinks the coefficient estimates towards zero. In other words, it discourages learning a more complex or flexible model to avoid the risk of overfitting. It reduces the variance of the model, without a substantial increase in its bias.

Uplatz

Machine Learning Interview Questions

Do you think that treating a categorical variable as a continuous variable would result in a better predictive model?

For a better predictive model, the categorical variable can be considered as a continuous variable only when the variable is ordinal in nature.

How is machine learning used in day-to-day life?

Most of the people are already using machine learning in their everyday life. Assume that you are engaging with the internet, you are actually expressing your preferences, likes, dislikes through your searches. All these things are picked up by cookies coming on your computer, from this, the behavior of a user is evaluated.

Describe dimension reduction in machine learning.

Dimension reduction is the process which is used to reduce the number of random variables under considerations.

Dimension reduction can be divided into feature selection and extraction.

Uplatz

Machine Learning Interview Questions

What do you understand by Reinforcement Learning technique?

Reinforcement learning is an algorithm technique used in Machine Learning. It involves an agent that interacts with its environment by producing actions & discovering errors or rewards. Reinforcement learning is employed by different software and machines to search for the best suitable behavior or path it should follow in a specific situation.

What is the trade-off between bias and variance?

Both bias and variance are errors. Bias is an error due to erroneous or overly simplistic assumptions in the learning algorithm. It can lead to the model under-fitting the data, making it hard to have high predictive accuracy and generalize the knowledge from the training set to the test set.

Variance is an error due to too much complexity in the learning algorithm. It leads to the algorithm being highly sensitive to high degrees of variation in the training data, which can lead the model to overfit the data.

Uplatz

Machine Learning Interview Questions

What do you mean by ensemble learning?

Numerous models, such as classifiers are strategically made and combined to solve a specific computational program which is known as ensemble learning. The ensemble methods are also known as committee-based learning or learning multiple classifier systems. It trains various hypotheses to fix the same issue. One of the most suitable examples of ensemble modeling is the random forest trees where several decision trees are used to predict outcomes.

What do you understand by Decision Tree in Machine Learning?

Decision Trees can be defined as the Supervised Machine Learning, where the data is continuously split according to a certain parameter. It builds classification or regression models as similar as a tree structure, with datasets broken up into ever smaller subsets while developing the decision tree. The tree can be defined by two entities, namely decision nodes, and leaves.

Describe 'Training set' and 'training Test'.

In various areas of information of machine learning, a set of data is used to discover the potentially predictive relationship, which is known as 'Training Set'. The training set is an example that is given to the learner. Besides, the 'Test set' is used to test the accuracy of the hypotheses generated by the learner. It is the set of instances held back from the learner.

Uplatz

Machine Learning Interview Questions

What do you mean by Genetic Programming?

Genetic Programming (GP) is almost similar to an Evolutionary Algorithm, a subset of machine learning. Genetic programming software systems implement an algorithm that uses random mutation, a fitness function, crossover, and multiple generations of evolution to resolve a user-defined task.

Describe the classifier in machine learning.

A classifier is a case of a hypothesis or discrete-valued function which is used to assign class labels to particular data points. It is a system that inputs a vector of discrete or continuous feature values and outputs a single discrete value, the class.

What is SVM in machine learning? What are the classification methods that SVM can handle?

SVM stands for Support Vector Machine.

The classification methods that SVM can handle are:

Combining binary classifiers

Modifying binary to incorporate multiclass learning

Uplatz

Machine Learning Interview Questions

What are the Recommended Systems?

Recommended System is a sub-directory of information filtering systems. It predicts the preferences or rankings offered by a user to a product. According to the preferences, it provides similar recommendations to a user. Recommendation systems are widely used in movies, news, research articles, products, social tips, music, etc.

Why do we need to convert categorical variables into factor? Which functions are used to perform the conversion?

Most Machine learning algorithms require number as input. That is why we convert categorical values into factors to get numerical values. We also don't have to deal with dummy variables.

The functions factor() and as.factor() are used to convert variables into factors.

Uplatz

Machine Learning Interview Questions

Uplatz

Machine Learning Interview Questions

Uplatz

Machine Learning Interview Questions

Uplatz

Machine Learning Interview Questions

Uplatz

Machine Learning Interview Questions

Uplatz

Machine Learning Interview Questions

Uplatz

Machine Learning Interview Questions

Uplatz